

Coursework 2

This coursework consists of two parts, (1) an implementation part where you are required to develop a web app, and (2) a theoretical part where you are required to write a critical report discussing some of the web technologies studied in the course.

Assessment Brief

1. Module number	SET11105, SET11505
2. Module title	Programming for the Web
3. Module leader	Dr Amjad Ullah
4. Tutor with responsibility for this Assessment	Dr Amjad Ullah, Dr Yanchao Yu
5. Assessment	Practical Assessment
6. Weighting	75% of module assessment
7. Size for assessment	Approx. 15 hours for Part 1 (webapp development) and approx. 15 hours for Part 2 (critical report).
8. Deadline of submission and Demonstration	Submission: Friday 15th April by 15:00 Demonstration: Will be held during teaching sessions of Week 14 (i.e., the week after submission). Schedule will be announced nearer the time.
9. Arrangements for submission	This coursework is assessed in two parts, a code development part and a critical report. The details on how to submit can be found at the end of this document under the section titled “Coursework submission”.
10. Assessment Regulations	All assessments are subject to the University Regulations.
11. The requirements for the assessment	This coursework is based on the taught academic content of the module (all LOs).
13. Return of work and feedback	Marks and feedback will normally be communicated via Moodle within 3 weeks of submission.
14. Assessment criteria	The implementation code and report will be marked against the criteria laid out in the respective Marking Schedule.

Part 1

This part refers to the creation of a full (font-end and back-end) Web App meeting the requirements outlined in the following scenario. You should do this part using the technologies taught in the module (e.g., ASP.NET MVC with C# and Razor, Entity Framework, AJAX, JSON, HTML5, Bootstrap, jQuery) and you should re-use the front-end created for Coursework 1. The scenario is exactly the same as in **Coursework 1**.

Scenario – Vote with Your Wallet

You are approached by an enthusiastic friend looking to create a Web App enabling people to publicly express their objection to creative tax avoidance, unfair employment practices, racism, disregard of environmental impact and other unethical business and political practices. The idea is that users will be able to add their signatures under a cause, which can be created by other members, supporting with the cause and pledging not to do business or otherwise deal with the misbehaving entity highlighted in the cause.

Although you like this idea, and you can foresee it getting some traction, you also know that there is no way you're going to be paid for this work, so you agree to undertake it for free, thinking that you'll at least gain some experience and reputation. However, you're hoping that it won't take that much of your time to produce something that is usable and looks cool.

In the next meeting, the friend explains a number of features wanted for the website, which you have ranked in increasing order of the perceived difficulty as follows:

1. Ability to create Member accounts
2. Members can create new Causes
3. Ability to electronically sign under Causes using a Member account
4. There is a special "Admin" account which can delete Causes
5. A list of names who have signed a Cause should be visible to anyone
6. A counter of how many people have signed a Cause should be visible to anyone
7. The website must be usable from an Internet-capable mobile phone
8. Ability to see signature counts and actual new signatures as they happen
9. Ability to share causes in social media like Facebook, Twitter, Reddit etc

10. Your friend says “It would be great if the Web App was actually a Smartphone App”

In turn, you explain that you’ll be using HTML5, jQuery, Bootstrap, ASP.NET MVC and a Database and you both agree that you should begin by making the Web App, starting with the user interface and then the server-side code.

Tasks

For this part of the coursework, you are required to:

- Develop the software to facilitate the scenario (front-end and back-end). You can re-use the code developed in Coursework 1
- To get higher marks, you must use all the following technologies
 - i Back-end: ASP.NET MVC in C# and the Razor view engine, Entity Framework with any database engine
 - ii Front-end: AJAX, JSON, jQuery, HTML5, Bootstrap
- You are expected to use the technologies above (i.e., you can’t do it in PHP). Otherwise, you will lose the related marks (see marking schedule).
- Look at the marking schedule detailing the evaluation criteria. Ensure you are meeting these, within the context of the scenario.
- Submit a text document containing the Controllers, Views, Models, CSS and Scripts that **you** have developed (that is **your code** only – there is no need to submit code made by others, e.g., the jQuery file).

Notes on Plagiarism

The following guidelines explain how the Academic Conduct Regulations will be applied in this coursework, and what you’re allowed to do in terms of using code from various sources.

1. You can re-use code from Coursework 1 and you do not need to reference it at all.
2. You can use code from sources found on the internet (e.g. StackOverflow) but these **must** be referenced in the form of comments in the code. You should include the link where the code is found and, if known, the author’s name or username/nickname.
3. You can use code from other Edinburgh Napier University students, either current or previous, in which case you should reference it with their full name and student ID.

Failing to do the above, even if you have amended the code, may result in an investigation for academic misconduct. In the case of (3) **all** students involved, including the original author of the original code, will be investigated. Note also that marks will be awarded only for your own code.

Marking schedule

This part worth 12 marks that accounts for 25% of the module. Your work for this part will be marked against the following specification. Each element below is marked as follows:

Criterion not met/Question not answered:

0 marks

Criterion partially met/Question not answered:

1 mark

Criterion met very well/Question answered well:

2 marks

No	Item	Mark (0,1,2)
1	Use of ASP.NET MVC with C# - Properly coded, well-structured and commented - Can answer related question	
2	Use of Entity Framework - The database populates on every object creation - Can answer related question	
3	Use of jQuery, AJAX and JSON - Properly coded, well-structured and commented - Live updates for both new signatures and counter	
4	Ability to create member accounts which can: - Create new Causes - Sign Causes - Admin account which can delete Causes - Signatures under Causes are visible to everyone - Can answer related question	
5	The following features are implemented: - A counter of how many people have signed a Cause - Counter and signatures update as they happen - The website must be usable from a mobile phone - Can answer related question	
6	Ability to share causes in social media - Either of Facebook, Twitter, Reddit or any other (total of two for full marks)	

Maximum of 12 marks are available.

Marker's Notes

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Part 2

In this part, you are asked to write a critical report discussing the following sets of technologies:

- ASP.NET MVC vs PHP and other Web Application Frameworks
- Bootstrap vs plain CSS
- jQuery vs plain JavaScript
- Entity Framework vs plain Database connection

This part is not related with the Web App done in Part 1 and will follow its own marking schedule (see below).

Tasks

For this part of the coursework, you're required to do your own independent investigation on the listed technologies as well as what makes web technologies successful, and then critically evaluate the related technologies within 1500 words.

Marking schedule

The report is worth 24 marks and 50% of the module. These marks will be shared as such:

- 4 marks for the overall presentation of the report, including:
 - Organisation with appropriate headers/sections/subsections
 - Inclusion of Introduction, Conclusions and References
 - Following of academic conventions when citing and referencing sources
- 20 marks for critical evaluation of said technologies (5 for each set)
 - Your view on the technologies
 - How the technologies compare between each other
 - Any problems with specific technologies
 - Why are these technologies are important and/or popular

Note: the report should be **no more than** 1500 words long (excluding references). Parts of the report exceeding this limit **will not be read** by the markers and will not be awarded any marks!

Submission instruction

You are required to submit one zipped file (.zip or .rar or .7z) called cw_<your matric number> on Moodle. The content of this zip file should include the following two documents:

- A text (word/PDF) document containing the source code of **Part 1** in a nicely and understandable format. Please note, as earlier mentioned, this document should contain the Controllers, Views, Models, CSS and Scripts that ***you*** have developed (that is ***your code*** only – there is no need to submit code made by others, e.g., the jQuery file). You are advised to keep your own copy of the assessment.
- Another text (word/PDF) document containing your critical report for **Part 2**.